

2.

$$J(s, u) = \frac{1}{2} (S_N - S_{goal})^T Q_N (S_N - S_{goal}) + \frac{1}{2} \sum_{k=0}^{N-1} \left[(S_k - S_{goal})^T Q (S_k - S_{goal}) + u_k^T R u_k \right]$$

$$\hat{S}_N = S_N - \bar{S}_N$$

$$\hat{S}_k = S_k - \bar{S}_k$$

$$\hat{u}_k = u_k - \bar{u}_k$$

$$\therefore S_N - S_{goal} = \hat{S}_N + \bar{S}_N - S_{goal}$$

$$S_k - S_{goal} = \hat{S}_k + \bar{S}_k - S_{goal}$$

$$u_k = \hat{u}_k + \bar{u}_k$$

$$\therefore J(s, u) = \frac{1}{2} (\hat{S}_N + \bar{S}_N - S_{goal})^T Q_N (\hat{S}_N + \bar{S}_N - S_{goal}) + \frac{1}{2} \sum_{k=0}^{N-1} \left[(\hat{S}_k + \bar{S}_k - S_{goal})^T Q (\hat{S}_k + \bar{S}_k - S_{goal}) + (\hat{u}_k + \bar{u}_k)^T R (\hat{u}_k + \bar{u}_k) \right]$$

$$\therefore (\bar{S}_N - S_{goal})^T Q_N \hat{S}_N \iff q_N^T \hat{S}_N$$

$$(\bar{S}_k - S_{goal})^T Q \hat{S}_k \iff q_k^T \hat{S}_k$$

$$\bar{u}_k R \hat{u}_k \iff r_k^T \hat{u}_k$$